

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10 – SUB-STRAND 3.4 LINEAR MOTION: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the unit. Driving question: How can mathematics help us fully describe, graph, and predict the motion of objects moving in a straight line, and why does direction always matter? Answer every section in the space under the prompt. Use full sentences, show every step of any calculation, and include units. Refer back to the Timer Race Puzzle, the Driving Question Board and your cumulative motion model from Lessons 1 to 10 where you can. Sketch graphs neatly and label both axes. You may use the formulae $v = s/t$ and $a = (v - u)/t$, and what you learned about graphs.

Part 1: Solving the Timer Race Puzzle

In the Timer Race, Group A walked a straight line, Group B walked a zigzag path and Group C walked the zigzag path and then retraced their steps back to a midpoint. Group C covered a similar total distance to the others, yet ended up in a completely different place. (a) Explain why this happened using the terms distance, displacement, scalar and vector. (b) Eliud Kipchoge runs 42 km in a marathon yet ends only a few hundred metres from where he started. Use your explanation from (a) to say what this tells us about his distance and his displacement. (c) Why is knowing the distance alone not enough to say where a traveller ends up?

(a) Distance is the total length of the path travelled and is a scalar, so it has size only. Displacement is the straight-line change in position from start to finish and is a vector, so it has size and direction. Group C walked a long path, so their distance was similar to the others. But they retraced their steps, which means they moved in opposite directions and part of their progress was cancelled. Their displacement was therefore smaller and their final position different. (b) Kipchoge's distance is 42 km because that is the length of the path he ran. His displacement is only a few hundred metres because it measures just the straight-line change from his start to his finish, and his route brings him back near where he began. (c) Distance gives only the size of the journey and says nothing about direction. Two travellers can have the same distance and the same time but end in different places, so we need displacement, which includes direction, to know where someone ends up.

Part 2: Speed, velocity and the matatu speedometer

(a) A matatu driver says, 'My speedometer shows 60 km/h.' Explain why this reading cannot tell you which direction the matatu is heading. (b) Write the formula for speed and the formula for velocity, and say which one is a vector. (c) A matatu makes a return trip along Thika Road and ends at its starting point. What are its displacement and its average velocity for the whole trip? Is its average speed also zero? Explain.	(a) A speedometer shows only how fast the matatu is going, which is its speed, a scalar. Speed has no direction, so 60 km/h could be towards Nairobi or away from it. (b) Speed = distance ÷ time. Velocity = displacement ÷ time. Velocity is the vector because it has a direction, shown by a sign or a stated direction. (c) The matatu finishes where it started, so its displacement is zero. Average velocity = displacement ÷ time = 0 ÷ time = 0, so its average velocity is zero. Its average speed is not zero because it travelled a real distance on the road, and speed uses total distance, which is greater than zero.
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Part 3: Reading motion from a displacement–time graph

Walker C in the Timer Race walked away from the start, then retraced steps back towards the start. (a) Describe in words what the displacement–time graph for Walker C would look like. Say what a positive gradient and a negative gradient each mean. (b) What does a horizontal section of a displacement–time graph tell you? (c) Explain why this graph shows direction, but a speedometer or a distance value does not.	(a) The graph first rises, with a positive gradient, as Walker C moves away from the starting point. Then it slopes downwards, with a negative gradient, as she retraces her steps. The gradient of a displacement–time graph is the velocity ($\Delta s \div \Delta t$). A positive gradient means motion away from the reference point. A negative gradient means motion back towards or past the reference point. The steeper the line, the greater the speed. (b) A horizontal section has zero gradient, so the velocity is zero and the object is stationary. (c) The gradient is velocity, which is a vector, so its sign shows direction as well as size. A negative gradient shows the return direction. A speedometer or distance value is a scalar and has no sign to show direction.
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Part 4: Acceleration and the boda-boda stopping distance

A boda-boda is travelling at 30 km/h and brakes uniformly until it stops in 5 seconds. (a) Convert 30 km/h to m/s, giving your answer to 2 decimal places. (b) Calculate the acceleration using $a = (v - u) / t$. Explain what the sign of your	(a) $30 \text{ km/h} = 30\,000 \text{ m} \div 3600 \text{ s} = 8.33 \text{ m/s}$. (b) $u = 8.33 \text{ m/s}$, $v = 0 \text{ m/s}$, $t = 5 \text{ s}$. $a = (v - u) / t = (0 - 8.33) \div 5 = -1.67 \text{ m/s}^2$. The negative sign means the velocity is decreasing, so the boda-boda is decelerating (slowing down). (c) The graph is a straight line sloping downwards from 8.33 m/s at $t = 0 \text{ s}$ to 0 m/s at $t = 5 \text{ s}$, with velocity on the vertical axis and time on the horizontal axis. The shape under the line is a triangle. Area = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 5 \times 8.33 = 20.8$, which is about 21 m. The area under a velocity–time graph equals displacement, so the stopping distance is approximately 21 m forward. (d) Acceleration is the rate of change of velocity. The velocity is getting smaller while the boda-boda still moves forward, so $(v - u)$ is negative. A negative acceleration here means slowing down, not moving backwards.
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answer means. (c) Sketch the velocity–time graph for the braking. Use the area under the graph to find the stopping distance, and give your answer to the nearest metre. (d) Why does the boda-boda have a negative acceleration even though it is still moving forward?	
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Part 5: Relative speed and the final answer to the driving question	
Two matatus travel on the Nairobi–Mombasa highway. (a) State the rule for the relative speed of the two matatus when they travel in the same direction, and the rule when they travel in opposite directions. Use v_1 and v_2 for their speeds. (b) Explain why direction decides which rule you use. (c) Using the Timer Race Puzzle, the marathon and at least two kinds of graph or formula from this unit, write a short answer to the driving question: How can mathematics help us fully describe, graph and predict motion in a straight line, and why does direction always matter?	(a) Same direction: relative speed = $v_1 - v_2$. Opposite directions: relative speed = $v_1 + v_2$. (b) When the matatus move the same way, one matatu's motion partly cancels the other's, so we subtract. When they move towards each other, their motions add together to close the gap, so we add. Direction decides whether the velocities combine by subtraction or addition. (c) The Timer Race Puzzle showed that the same distance and the same time can leave travellers in different places, so distance and speed alone are not enough. Mathematics gives us displacement, velocity and acceleration, which are vectors that carry direction in their sign. The formulae $v = s/t$ and $a = (v - u)/t$ let us calculate them. A displacement–time graph has a gradient equal to velocity, and a negative gradient shows motion back towards the start. A velocity–time graph has a gradient equal to acceleration, and the area under it equals displacement, so we can find the boda-boda's stopping distance and predict whether an overtake is safe. This is why Kipchoge can run 42 km yet have a small displacement, and why a matatu's speedometer cannot tell you where it is headed. Direction always matters because without it we can describe how much motion there was, but not where the object ends up.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Distinguishing scalars and vectors (distance/displacement, speed/velocity)	Defines all four terms correctly, labels each as scalar or vector, and uses them accurately to explain the Timer Race Puzzle and Kipchoge's marathon.	Defines most terms correctly and uses them to explain the puzzle, with minor gaps in scalar/vector labelling.	Mixes up distance and displacement or speed and velocity, or does not link them to the puzzle.
Calculations and units	Converts units correctly, applies $a = (v - u)/t$ and the area under the graph	Applies the formulae correctly with small slips in units, rounding or sign, and	Uses the wrong formula, omits units or

	correctly, gives the sign and units, and shows every step.	shows most steps.	the sign, or does not show working.
Interpreting graphs	Explains that the gradient of a displacement–time graph is velocity and the gradient of a velocity–time graph is acceleration, interprets negative and zero gradients, and uses area as displacement.	Interprets gradients correctly but is less clear on negative gradients or on area as displacement.	Describes the graph shape only, or confuses displacement–time and velocity–time graphs.
Explaining why direction matters	Consistently links the sign of velocity, acceleration and gradient to direction, and includes relative speed rules with a clear reason for adding or subtracting.	States that direction matters and gives some correct links to signs or relative speed.	Says direction matters without evidence or explanation.
Use of evidence and the driving question	Draws on the Timer Race Puzzle, Kipchoge, the matatu and the boda-boda to give a complete, well-organised answer to the driving question.	Uses some unit examples to answer the driving question with reasonable organisation.	Gives a general answer with few or no examples from the unit.

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